
❏ What is Virtualization?

Virtualization is the process of creating a **virtual version** of something—such as an operating system (OS), a server, a storage device, or network resources—using **software** to simulate hardware functionality.

Key benefits:

- Better resource utilization
 - Easier management and provisioning
 - Cost savings
 - Isolation and portability
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☒ Types of Virtualization

1. Type 1 Hypervisor (Bare-metal)

- Installed directly on physical hardware.
- Offers **better performance** and security.
- Common in **data centers** and enterprise environments.

Examples:

- VMware ESXi
- Microsoft Hyper-V (bare-metal)
- Xen

Advantages:

- No dependency on host OS
- Direct access to hardware resources

2. Type 2 Hypervisor (Hosted)

- Installed on top of a host operating system.
- Easier to set up; often used in **development or testing** environments.

Examples:

- VMware Workstation

- Oracle VirtualBox
- Parallels

Disadvantages:

- Performance overhead due to host OS
- Less secure compared to Type 1

☒ Key Virtualization Concepts

☒ Hardware Virtualization

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Uses a hypervisor to abstract and allocate hardware to virtual machines (VMs).

- The hypervisor simulates virtual hardware (CPU, RAM, NICs, storage).

☒ Para-Virtualization

- Guest OS is **aware it is being virtualized**.
- It interacts with the hypervisor using special APIs instead of full hardware emulation.
- **More efficient** than full virtualization but **requires modified OS**.

Examples:

- Xen supports both para- and full virtualization.

☒ Cloning

- Creating an **exact copy** of a virtual machine.
- Useful for quickly deploying new VMs with the same configuration.
- Two types:
 - **Full Clone:** Independent VM
 - **Linked Clone:** Shares base disk (saves space)

☒ Snapshot

- Captures the **state of a VM** at a specific point in time.

- Used for backup or testing.
- Can **revert** to a snapshot if an update or change causes issues.

Note: Snapshots are not backups—they rely on the base VM and should be used with caution in production.

☒ Template

- A **read-only master image** used to create new virtual machines.
- Pre-installed OS and configuration
- Streamlines deployment and ensures consistency

☒ Operating System Virtualization

- Virtualizes at the OS level, allowing multiple **isolated user-space instances** on the same kernel.

Examples:

- Linux Containers (LXC)
- Docker (uses containerization, a form of OS virtualization)

Benefits:

- Lightweight compared to hypervisor-based virtualization
- Fast startup and better performance
- Ideal for microservices and cloud-native apps

❏ Cluster Architecture

A **cluster** is a group of interconnected computers that work together as a single system to ensure **availability**, **scalability**, and **reliability**.

Types of clusters:

1.
High-Availability (HA): Ensures services are available even if one node fails.
2.
Load-Balancing: Distributes workloads across nodes.
3.
High-Performance Computing (HPC): Combines resources for intense computing.

Core Components:

- Nodes (physical or virtual)

- Cluster Management Software (e.g., vCenter, Kubernetes, Pacemaker)
- Shared Storage (e.g., SAN, NAS)

❏ Cluster Requirements

To set up a virtualization cluster, you need:

Requirement	Description
Shared Storage	Required for VM migration and high availability.
Consistent Hardware	Nodes should have similar specs for load balancing.
Networking	Redundant NICs and proper VLAN configuration.
Cluster Software	Tools like vCenter, Proxmox, or Kubernetes.
Fencing	Mechanism to isolate a failed node (e.g., IPMI, power fencing).
Heartbeat Mechanism	Detects node failures by exchanging regular status updates.

Benefits of clustering in virtualization:

- VM High Availability
- Live Migration (vMotion)
- Improved Fault Tolerance
- Scalability and resource pooling